

UNIVERSITY OF EDINBURGH
SCHOOL OF INFORMATICS

INFR11287 ADVANCED TOPICS IN NATURAL LANGUAGE
PROCESSING

Mock Examination Paper 1

Retrieval, Question Answering, Data, Translation, and Evaluation

Time allowed: 2 hours

INSTRUCTIONS TO CANDIDATES

1. This is a practice paper, not an official examination paper.
2. Answer all questions. Different questions may have different numbers of total marks.
3. The paper is worth 50 marks in total.
4. Notes, calculators, and dictionaries are not permitted in the real examination.
5. Please answer concisely. Unless otherwise stated, do not write more than one or two sentences per mark assigned.

1. You are working with a public health organisation that wants to build a machine translation and data pipeline for English, Swahili, and Yoruba. The organisation has a small amount of English-Swahili and English-Yoruba parallel text, a larger collection of monolingual Swahili and Yoruba health pages, and a very large general web crawl. It wants to support reliable translation of health advice for local clinics.
- (a) Describe the major steps you would take to create additional parallel English-Yoruba data from the web crawl. Include how you would identify candidate documents and how you would decide whether sentence pairs are parallel. [3 marks]
 - (b) The first English-Yoruba translation model trained on the small parallel corpus performs poorly. Explain two ways in which the monolingual Yoruba data could be used to improve the English to Yoruba system. For each method, note one quality-control step. [4 marks]
 - (c) The organisation also wants to select Swahili medical text from the large general crawl. Explain the difference between quality-based, diversity-based, and importance-based data selection, giving one suitable example of each for this setting. [3 marks]
 - (d) The team considers initialising the Yoruba model from a high-resource German-English translation model and training a shared BPE vocabulary over English, German, and Yoruba. State one advantage and one possible problem with this multilingual BPE strategy. [2 marks]
 - (e) Describe an evaluation plan for the final health translation system. Your answer should include at least one automatic metric, one human evaluation component, and one possible ethical or social harm if the model is deployed without adequate evaluation. [3 marks]

2. Alex is building an open-domain question answering system for an international fact-checking service. The system should answer questions using Wikipedia, official reports, and local-language news sources. Alex proposes to use a retrieval augmented generation system rather than simply asking a large language model that was pre-trained on web data.
- (a) Explain one advantage and two disadvantages of using RAG rather than relying only on the parametric knowledge of the language model. [4 marks]
 - (b) Alex has to choose between sparse retrieval, dense retrieval, and a hybrid retriever. Compare sparse and dense retrieval in this setting, and explain why a hybrid system with reranking might be useful. [3 marks]
 - (c) For Arabic-speaking users, the first prototype translates the Arabic question into English, retrieves only English documents, generates an English answer, and translates the answer back into Arabic. Describe two problems with the quality of answers this pipeline might produce, and one ethical risk. [3 marks]
 - (d) Alex considers two alternatives: indexing non-English Wikipedia directly, or translating all non-English documents into English before indexing. Give one advantage and one disadvantage of each approach. [3 marks]
 - (e) Describe how you would evaluate this QA system. Your answer should include retrieval evaluation, answer evaluation, and at least one check for contamination or benchmark saturation. [2 marks]

3. The following are multiple choice questions. For each question, choose the answer that is the most correct. Each question is worth 2 marks. [20 marks]

- (a) Compared to ROUGE, BERTScore can improve automatic evaluation of summaries because:
 - A. It never requires reference summaries.
 - B. It can give credit for semantic similarity without exact lexical overlap.
 - C. It is guaranteed to detect hallucinations.
 - D. It is a human evaluation protocol rather than an automatic metric.
- (b) Which statement about the Natural Questions dataset is true?
 - A. It consists only of manually invented questions with no connection to real search queries.
 - B. It includes naturally occurring Google Search questions with Wikipedia evidence.
 - C. It only contains yes/no questions.
 - D. It is a summarisation benchmark rather than a QA benchmark.
- (c) Moore-Lewis data selection is best described as selecting examples that:
 - A. Have the lowest probability under every language model.
 - B. Are maximally different from the target domain.
 - C. Are explained better by an in-domain language model than by a general-domain language model.
 - D. Have the highest number of unique tokens, regardless of domain.
- (d) The usual input to a reference-based COMET evaluation model includes:
 - A. Only the translated sentence.
 - B. Only the source sentence and the model parameters.
 - C. The source sentence, the system translation, and a reference translation.
 - D. The training loss and validation perplexity of the MT system.
- (e) Benchmark contamination is a problem because it can:
 - A. Make test scores appear better than the model's true generalisation.
 - B. Prevent any model from using chain-of-thought prompting.
 - C. Only affect models trained with reinforcement learning.
 - D. Always reduce evaluation scores.
- (f) In Fusion-in-Decoder style open-domain QA:
 - A. Retrieval is not used.
 - B. Candidate passages are encoded and the decoder fuses information while generating the answer.
 - C. The answer must be a span copied from a single passage.
 - D. All facts must be stored in the model parameters.
- (g) The "curse of multilinguality" refers to the observation that:
 - A. Multilingual models can never transfer between languages.
 - B. Adding more languages always improves every language equally.
 - C. Fixed model capacity and unbalanced data can cause interference or weaker performance for some languages.
 - D. Evaluation in English is impossible for multilingual models.
- (h) LM Arena style evaluation usually relies on:
 - A. Exact string matching against a gold answer.
 - B. Pairwise user preferences converted into a rating such as Elo.
 - C. Calculating token-level perplexity on Wikipedia.

- D. Measuring only training speed.
- (i) Backtranslation is useful in low-resource MT primarily because:
 - A. It removes the need for any target-language data.
 - B. It guarantees perfect synthetic source sentences.
 - C. It creates pseudo-parallel data whose target side can come from real monolingual target-language text.
 - D. It evaluates translation quality without a reference.
- (j) In the Word Embedding Association Test with male and female names, science terms, and family terms, the test is mainly measuring:
 - A. Whether all names are equally frequent in the corpus.
 - B. Whether the embedding dimension is large enough.
 - C. Whether one target group is more strongly associated with one attribute set than another target group is.
 - D. Whether an MT system preserves named entities.